

Jairs

29 July 2026

A Jai-inspired systems language in Rust. Incremental salsa front end, lossless CST, typed SSA mid-end, two execution engines held byte-identical by a differential harness.

6/6 gates green 892 tests ADR-0047 latest W1: 1 feature left uncommitted

TESTS

892

workspace, all passing

CORPUS

110

jr files, both engines

ADRS

47

0001 to 0047, immutable

DIAGNOSTICS

81

codes, E0246 next free

EDITOR CHECKS

98

Neovim, verified not gated

The vertical slice, and what it costs to be honest about it

EXIT CRITERIA

DONE hello.jr runs in the VM
DONE compiles to a native binary
DONE rustc-grade diagnostics
DONE formatter round-trips the corpus
DONE editor integration (Neovim)
OPEN verified Linux x86-64 CI run

The last one needs a push. Configured, never run, so it is a decision rather than a technical gap.

HOW CORRECTNESS IS ESTABLISHED

Both engines run every corpus program and must agree, on output and on trap wording, from one shared formatter.
MIR snapshots pin the exact instruction sequence for all 110 files. That caught a whole wave's regression this session.
Every fix is verified by reverting it and watching the test fail. A test that passes without its fix is worse than none.

The language today

WORKS, END TO END

Full integer tower, bool, string, pointers
float32 and float64, plain IEEE-754, no traps
struct, nominal, one level
union, nominal, untagged, fields at offset 0
enum and enum_flags, namespaced members
Fixed arrays: index, count, zeroed, bounds-checked
Views: buf slice, writes through to the array
cast and xx from context; bare dot-member
Trapping arithmetic and wrapping variants
Bitwise ops, non-C precedence, trapping shift
if, else, while, break, continue, return
Procedures, one return value
import, foreign, system_library
One trivial compile-time run, folded
Traps name their source line

ABSENT (WAVE)

pointer arithmetic, deliberately none
percent on floats, is_nan, math (W7)

a tagged variant type (W2)
explicit backing type
array literals
sub-slicing, equality on views
operator overloading, last of W1

for, defer, using (W2)
multiple returns, named args (W2)
polymorphs, macros (W5)
arbitrary run, RTTI, insert (W4)
context, allocators, backtraces (W3)

No error-handling model yet, ADR-0008 reserves the slot and nothing fills it. No GC and no RAII, which is a design value rather than a gap. A union is untagged, so reading a field other than the one last written reinterprets bits: documented in three places because it cannot be diagnosed.

Compiler internals, 17 crates

STAGE	STATE	NOTE
Lexer, parser, CST, typed AST	works	Hand-written, error-recovering, trivia-preserving
Formatter	works	Pure function over the CST; the corpus is canonical under it
HIR, name resolution, modules	works	Flat import merge; cycles legal
InternPool: types, values, layout	works	One layout computation and one integer evaluator, shared
Sema: signatures, checking	works	E0212 to E0244; no const-eval here, by design
MIR: typed SSA	works	Block parameters, not phis; explicit bounds check and zeroing
Mid-end	4 passes	Inline, store-to-load forwarding, const-prop, DCE
Const-eval	works	Runs MIR through the bytecode VM
VM: register bytecode, libffi	works	No JIT tier
Cranelift back end	works	Aggregate params; cannot return one yet
LLVM back end	not started	W8 owns it
Language server	12 caps	Diagnostics, hover, goto, completion, rename, actions, hints
Neovim integration	works	Runtimepath dir, no plugin manager; 98 checks
Driver	stub	Should consume the workspace notion that now exists

Roadmap: W1 nearly closed, W2 through W10 ahead

W1 Data

Numeric tower, wrapping ops, enum, enum_flags, union, arrays, views, cast, xx: all done. Operator overloading remains and needs its own ADR, being a name-resolution and dispatch feature that shares nothing with the rest.

1 left

W2 Flow and scope

for with it and it_index, labelled break, defer, using. Two deferred decisions come due: a bare dot-member in a switch, and a tagged variant type beside union.

next

W3 Runtime core

context as a hidden parameter, allocators, temporary storage, the bounds-check build setting ADR-0003 asked for, panics and backtraces.

not started

W4 Comptime

Arbitrary compile-time execution, RTTI and Type values, insert, code. PLAN section 5 names this the project's top risk: sema and comptime become mutually recursive.

not started

W5 Polymorphism

Polymorphic parameters, modify, bake_arguments, expand macros with hygiene, instantiation caching.

not started

W6 Metaprogram

Workspaces, the compiler message loop, build scripts replacing makefiles, note attributes.

not started

W7 Stdlib

Written in Jairs: String, dynamic array, hash table, Sort, Math, Random, File.

not started

W8 Performance

LLVM via inkwell for release builds, inliner maturity, struct-of-arrays, SIMD, parallel sema. Also owns the compile-throughput number.

not started

W9 Tooling depth

Semantic tokens are the one LSP capability still missing; the other eleven landed early.

not started

W10 Graphics, in Jairs

Window creation through foreign calls, Metal then Vulkan, immediate-mode 2D.

not started

What this session did, including the part that went wrong

NINE WAVES SHIPPED

ADR-0039 arrays and bounds checks, 0040 floats, 0041 enums, 0042 bitwise, 0043 enum_flags, 0044 views, 0045 unions, 0046 xx and bare members, 0047 imported enum members.
Test count 846 to 892. Corpus 78 to 110 files. Four silent miscompiles found and fixed: aggregate zeroing, address-of-parameter, union field aliasing in store-to-load forwarding, and a recovery arm computing 6 and 3 as 9.

ONE SERIOUS MISTAKE

I ran git checkout on an uncommitted file as a casual undo. Because nothing this session is committed, that restored the MIR builder from a HEAD seven waves old and destroyed about 900 lines.
Rebuilt from the ADRs and the MIR snapshots, which pin the exact expected output. Restoration is verified: the snapshot matches byte-for-byte across all 110 files, and the test count returned to its prior value exactly.

THE THING WORTH YOUR DECISION

Nine waves, 116 changed files, sit uncommitted in one working tree, because the standing rule is that commits need explicit authorisation. That rule is what turned one careless command into a seven-wave loss rather than a one-wave one. Committing each wave as it goes green would bound the damage from any future mistake to a single wave.
Recorded as a recommendation, not a change: the authorisation is yours to give.

Sources: PLAN.md sections 1.5 and 7, the ADR directory, and the six gates run today. Every number here was measured rather than carried forward: the test count from a full workspace test run, the corpus count from a file walk, the diagnostic codes from a grep over each crate's code registry.